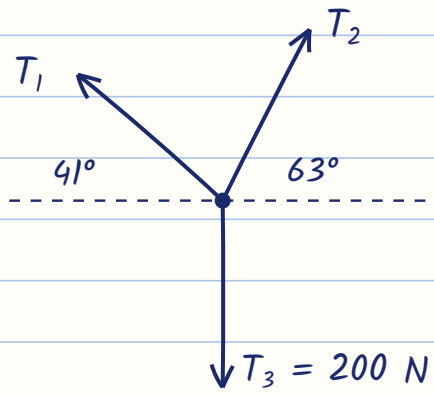
## 1) Traffic light - tension in each cable

Light is in equilibrium: $T_3 = w = 200$ N

FBD of the knot (angles from horizontal):




$\Sigma F_x = 0$:

$T_2 \cos 63° - T_1 \cos 41° = 0$

$\Sigma F_y = 0$:

$T_1 \sin 41° + T_2 \sin 63° = 200$

from x: $T_1 = T_2(\cos 63°/\cos 41°)$

$T_1 = T_2(.4540/.7547) = 0.6015\ T_2$

Plug into y: $0.6015\ T_2\ (0.6561) + T_2\ (0.8910) = 200$

$T_2\ (0.3946 + 0.8910) = 200$

$T_2\ (1.2856) = 200 \rightarrow T_2 = 155.6$ N

$T_1 = 0.6015\ (155.6) = 93.6$ N

check: $93.6(.6561) + 155.6(.8910) = 61.4 + 138.6 = 200$ ✓

**$T_1 = 93.6$ N  $T_2 = 156$ N  $T_3 = 200$ N**

## 2) Atwood's machine

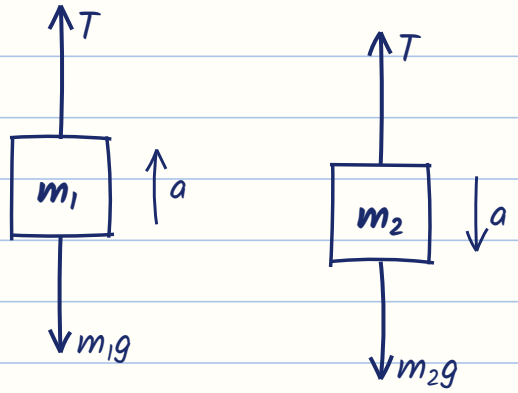
$m_2 > m_1$ so $m_2$ goes down, $m_1$ goes up, same $a$ and same $T$.

(a) $m_1$: $T - m_1 g = m_1 a$

$m_2$: $m_2 g - T = m_2 a$

add them ($T$ cancels):

$m_2 g - m_1 g = (m_1 + m_2)\, a$




$$\boxed{a = (m_2 - m_1)\, g \,/\, (m_1 + m_2)}$$

(b) $T = m_1(g + a) = m_1 g\,[1 + (m_2 - m_1)/(m_1 + m_2)]$

$T = m_1 g\,[\, 2m_2 \,/\, (m_1 + m_2)\,]$

$$\boxed{T = 2\, m_1\, m_2\, g \,/\, (m_1 + m_2)}$$

(c) $m_1 = 2.00$ kg, $m_2 = 4.00$ kg

$a = (4.00 - 2.00)(9.80) \,/\, (6.00) = 19.6 \,/\, 6.00 = 3.27$ m/s²

$T = 2(2.00)(4.00)(9.80) \,/\, 6.00 = 156.8 \,/\, 6.00 = 26.1$ N

check: $T = m_1(g + a) = 2.00(9.80 + 3.27) = 26.1$ N ✓

$$\boxed{a = 3.27 \text{ m/s}^2 \quad (m_2 \text{ down}) \qquad T = 26.1 \text{ N}}$$

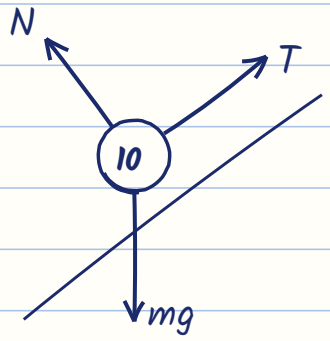
## 3) Two carts on inclines (frictionless)

Only gravity along each incline matters (N is ⊥ to motion).

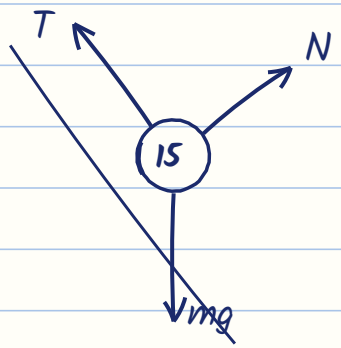
15 kg: $m_2 g \sin 53° = (15)(9.80)(0.7986) = 117.4$ N

10 kg: $m_1 g \sin 37° = (10)(9.80)(0.6018) = 59.0$ N

$117.4 > 59.0$ → 15 kg slides down, 10 kg is pulled up




10 kg on 37°




15 kg on 53°

Newton's 2nd law along each incline (+ = direction it moves):

10 kg: $T - m_1 g \sin 37° = m_1 a$

15 kg: $m_2 g \sin 53° - T = m_2 a$

add: $m_2 g \sin 53° - m_1 g \sin 37° = (m_1 + m_2)\, a$

$a = (117.4 - 59.0) / (10 + 15) = 58.4 / 25$

**$a = 2.34$ m/s²**

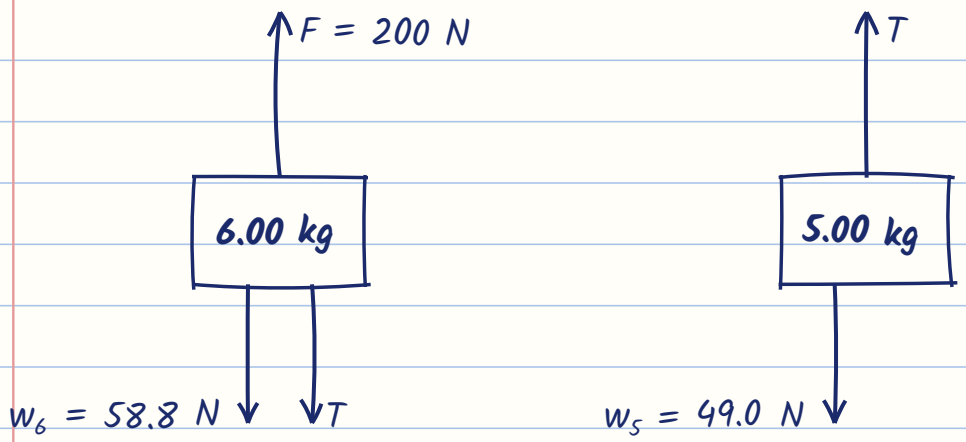
$T = m_1(a + g \sin 37°) = 10(2.34 + 5.90) = 82.4$ N

check: $15(7.83 - 2.34) = 82.4$ N ✓

**$T = 82.4$ N**

## 4) Two blocks pulled up by F = 200 N

(a) FBDs




6.00 kg: F (by whoever pulls up), T down (by the rope),
$w_6 = m_6 g$ down (by Earth)

5.00 kg: T up (by the rope), $w_5 = m_5 g$ down (by Earth)

(b) Whole system (T is internal, cancels):

$F - (m_6 + m_5)\, g = (m_6 + m_5)\, a$

$a = [200 - (11.0)(9.80)] / 11.0 = (200 - 107.8)/11.0 = 92.2/11.0$

**$a = 8.38$ m/s² (up)**

(c) 5.00 kg block alone:

$T - m_5 g = m_5 a \rightarrow T = m_5(g + a) = 5.00(9.80 + 8.38)$

check (6 kg): $200 - 90.9 - 58.8 = 50.3 = 6.00(8.38)$ ✓

**$T = 90.9$ N**